

NAPA COUNTY AP, CA, USA

WMO: 724955

Lat: 38.210N

Lon: 122.285W

Elev: 14

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 93227

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	29.6	32.1	17.5	13.3	51.7	22.2	16.7	49.2	25.0	55.9	21.7	55.2	2.1	0	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.0	91.1	65.9	86.4	64.9	82.4	64.0	68.4	86.1	66.8	82.9	65.4	79.7	10.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	81.6	74.3	60.4	78.5	72.7	59.2	75.2	71.0	32.6	86.2	31.2	82.9	30.2	79.9	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.3	19.1	17.5	DB	23.8	99.7	2.4	3.6	22.0	102.2	20.6	104.3	19.3	106.3	17.6	108.9
			WB	22.5	71.5	2.5	1.5	20.7	72.6	19.3	73.4	17.8	74.3	16.0	75.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.3	47.5	49.9	52.8	55.2	59.8	64.5	65.9	66.1	65.1	60.1	52.6	47.3	(6)
	DBStd	8.36	4.50	4.64	4.77	5.02	4.65	4.70	3.79	3.62	4.74	4.67	5.21	5.15	(7)
	HDD50	338	103	54	25	10	0	0	0	0	0	0	31	115	(8)
	HDD65	3076	543	423	380	296	176	62	30	26	54	165	371	550	(9)
	CDD50	2994	25	51	111	166	303	436	493	500	454	314	110	31	(10)
	CDD65	254	0	0	1	2	13	48	59	60	58	13	0	0	(11)
	CDH74	4025	1	4	34	128	332	758	666	713	926	436	26	1	(12)
	CDH80	1438	0	0	3	29	117	292	220	240	398	137	2	0	(13)
(14) Wind	WSAvg	7.5	6.3	6.4	6.9	8.1	9.1	9.8	10.0	9.1	7.3	5.8	5.3	6.2	(14)
(15) Precipitation	PrecAvg	26.4	5.4	4.6	3.7	1.5	0.7	0.2	0.0	0.1	0.3	1.5	3.7	4.8	(15)
	PrecMax	51.3	13.8	15.3	12.0	5.7	3.7	2.0	1.1	1.3	2.3	10.4	10.5	15.5	(16)
	PrecMin	10.5	0.4	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	(17)
	PrecStd	8.7	3.6	3.7	2.9	1.5	1.0	0.4	0.2	0.3	0.5	1.8	2.7	3.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.1	72.2	78.2	83.9	90.1	95.1	93.0	94.0	96.3	89.9	77.3	65.0	(19)
		MCWB	50.8	58.1	62.1	63.8	65.5	68.1	68.3	66.9	65.4	61.7	59.7	49.6	(20)
	2%	DB	62.4	67.2	72.7	77.5	82.6	88.0	85.6	86.2	90.5	83.8	72.4	61.8	(21)
		MCWB	52.1	55.5	59.1	61.3	63.1	65.6	67.0	66.5	64.7	60.1	57.0	52.6	(22)
	5%	DB	59.1	63.3	68.0	71.9	76.6	82.1	80.8	81.4	84.4	78.9	67.6	59.3	(23)
		MCWB	51.9	53.4	56.5	58.8	61.2	64.1	65.4	65.3	63.6	58.7	55.5	53.1	(24)
	10%	DB	56.4	60.1	64.0	67.1	72.0	77.4	76.8	77.2	78.6	73.9	63.7	57.1	(25)
		MCWB	51.4	52.8	55.0	57.0	59.8	62.6	63.9	64.0	62.9	58.1	54.6	51.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.3	60.1	64.0	66.3	68.7	70.2	70.2	69.6	69.1	65.9	62.9	59.0	(27)
		MCDB	60.4	67.4	75.5	80.4	86.3	91.2	89.3	87.8	87.3	80.9	73.5	60.8	(28)
	2%	WB	55.3	57.8	60.5	63.1	64.6	67.3	67.6	67.4	66.8	63.5	60.1	57.0	(29)
		MCDB	58.4	64.1	70.2	75.8	79.0	84.5	84.2	83.8	84.3	76.4	67.3	58.9	(30)
	5%	WB	53.8	55.9	58.2	60.2	62.4	65.1	65.8	65.8	65.3	61.8	58.3	55.2	(31)
		MCDB	56.4	61.0	65.4	69.7	74.5	79.6	79.9	80.0	80.6	73.3	63.9	57.1	(32)
	10%	WB	52.4	54.1	56.1	57.8	60.3	63.2	64.2	64.3	63.7	60.0	56.5	53.1	(33)
		MCDB	55.3	58.5	62.1	65.3	70.8	76.1	76.4	76.6	77.3	70.3	61.8	55.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.1	20.0	21.8	23.3	24.2	25.3	23.3	24.0	27.9	27.5	22.7	18.8	(35)
		MCDBR	23.8	27.3	29.8	32.8	34.2	35.0	31.1	32.7	39.1	37.4	29.8	21.7	(36)
		MCWBR	15.2	16.9	18.1	18.9	17.6	16.0	14.5	15.2	16.8	17.5	16.5	13.9	(37)
	5% WB	MCDBR	16.0	21.4	25.7	28.6	29.6	30.6	28.7	29.5	33.2	29.4	22.0	12.7	(38)
		MCWBR	11.6	14.2	16.7	17.6	16.5	14.7	13.7	14.0	15.3	15.4	14.1	9.9	(39)
(40) Clear-Sky Solar Irradiance	taub		0.320	0.321	0.334	0.356	0.360	0.357	0.361	0.345	0.343	0.321	0.318	0.321	(40)
	taud		2.548	2.548	2.501	2.411	2.430	2.444	2.429	2.468	2.477	2.576	2.560	2.546	(41)
	Ebn at Noon		274	290	295	292	292	292	289	292	288	285	273	265	(42)
	Edh at Noon		24	27	31	36	36	36	36	34	32	26	23	23	(43)
(44) All-Sky Solar Radiation	RadAvg		729	1034	1464	1932	2287	2528	2461	2192	1819	1304	874	639	(44)
	RadStd		116	134	146	161	141	108	75	75	63	89	61	102	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.74	N/A	-3.01	N/A	N/A	N/A	N/A	N/A	+254	N/A
Regional (47 neighbors)		+0.67	N/A	-2.07	N/A	N/A	N/A	N/A	-207	+239	N/A

Nomenclature: See separate page